

Phase 3 Report — DAFormer-Style Unsupervised Domain Adaptation

Starting point

Self-training unsupervised domain adaptation (Cityscapes → ACDC) began from the Phase 1 SegFormer/MiT-B4 checkpoint (73.18% mIoU) and the Phase 2 zero-shot baseline (49.02% mIoU average across fog, rain, snow and night). An initial run carried over Phase 1's learning rate directly and produced classic self-training collapse: ACDC validation mIoU peaked at epoch 3, then degraded below the source-only baseline as the student drifted toward its own noisy pseudo-labels. Lowering the learning rate to $1.5e-5$ slowed the drift but did not resolve the underlying cause.

BatchNorm contamination and its fix

Two subsequent runs excluded night from the pseudo-label training pool to isolate a suspected night-specific miscalibration (rising pseudo-label confidence alongside falling accuracy), but the whole model kept degrading regardless of which conditions it trained on. Diagnosis: BatchNorm running statistics, calibrated on Cityscapes in Phase 1, were being silently overwritten as source and target batches alternated during adaptation, corrupting the model's statistics for both domains at once. Freezing BatchNorm layers — keeping them on Phase 1's already-calibrated running statistics — fixed this: a real, mechanistically understood gain of roughly +1.1 points, modest next to the 10–20 point gains reported in the adaptation literature, but the first result that behaved correctly.

Adding feature-distance regularization and MIC

With the BatchNorm fix validated, the next run added ImageNet feature-distance regularization (penalizing the encoder for drifting away from a generalizable representation) and Masked Image Consistency (MIC), which forces the student to infer masked-out image regions from context alone — a mechanism specifically suited to low-visibility conditions. Night was re-included in pseudo-label training now that the corrupting mechanism was fixed. Across 20 epochs, ACDC mIoU rose from 49.83% (epoch 1) to a best of 53.27% (epoch 10), then gently declined to 51.54% by epoch 20 — a normal self-training peak-then-decline curve, with the peak checkpoint retained.

- Fog: 67.43% mIoU (+6.70 vs. source-only)
- Rain: 55.15% mIoU (+2.78 vs. source-only)
- Snow: 55.85% mIoU (+8.23 vs. source-only)
- Night: 34.64% mIoU (−0.71 vs. source-only)
- **Average: 53.27% mIoU (+4.25 over the 49.02% source-only baseline)**

Fog, rain and snow all improved cleanly, but night regressed slightly below its own zero-shot baseline — the one condition self-training alone could not fix.

Ablation study: isolating what actually helps

To confirm each component's contribution and plan a targeted fix for night, three 6-epoch ablations were run: baseline self-training alone, self-training with BatchNorm freeze, and the full combined configuration (BatchNorm freeze + feature-distance + MIC). Results confirmed each addition helps and the combined

setting is the strongest of the three — 48.09% → 49.71% → 51.50% average mIoU (Figure 1). All three were still climbing at 6 epochs, so these numbers understate each configuration's ceiling, but the ranking between them is stable.

Fixing night without sacrificing the rest

A follow-up run targeted specifically at improving night added rare-class source sampling, a tuned pseudo-label confidence threshold, a target-consistency loss term, and a stronger, night-tuned augmentation (heavier blur, motion blur and patch dropout). The first attempt at this configuration surfaced two bugs:

- **Loss instability** — the target-consistency loss used a KL-divergence reduction meant for flat classification labels, not per-pixel segmentation maps. On a 512×512, 19-class output this inflated that one loss term by roughly three orders of magnitude, dominating the gradient and destabilizing training after an early peak.
- **Cross-condition contamination** — the stronger augmentation intended specifically for night was being applied to every target image regardless of condition, measurably degrading fog, rain and snow in the first attempt even though nothing about those conditions had changed.

Both were corrected: the consistency loss is now averaged per pixel so it sits on the same scale as the other loss terms, and the heavier augmentation is now applied only to images actually drawn from the night split, identified from their file path.

Final result: 54.19% average mIoU

With both fixes in place, the final tuned run (BatchNorm freeze, feature-distance regularization, MIC, rare-class source sampling, and condition-aware night augmentation) trained cleanly for the full 20 epochs — loss stayed stable throughout (0.84–0.97, versus the earlier attempt's spike into the thousands) — and peaked at epoch 9 (Figure 4):

- Fog: 67.4% mIoU
- Rain: 56.6% mIoU (+1.5 over the first combined run)
- Snow: 57.8% mIoU (+1.9 over the first combined run)
- Night: 34.9% mIoU (+0.3 over the first combined run, though still marginally below the 35.35% zero-shot baseline)
- **Average: 54.19% mIoU (+0.92 over the first combined run, +5.17 over the 49.02% source-only baseline)**

This is the best result obtained across every condition at once: fog, rain and snow all match or improve on the first combined run, while night narrows most of the gap to its zero-shot ceiling without the earlier tuning attempt's collapse in the other three conditions. Night remains the hardest condition and the one place adaptation has not yet clearly beaten doing nothing — the natural target for any further work.

Results — Figures

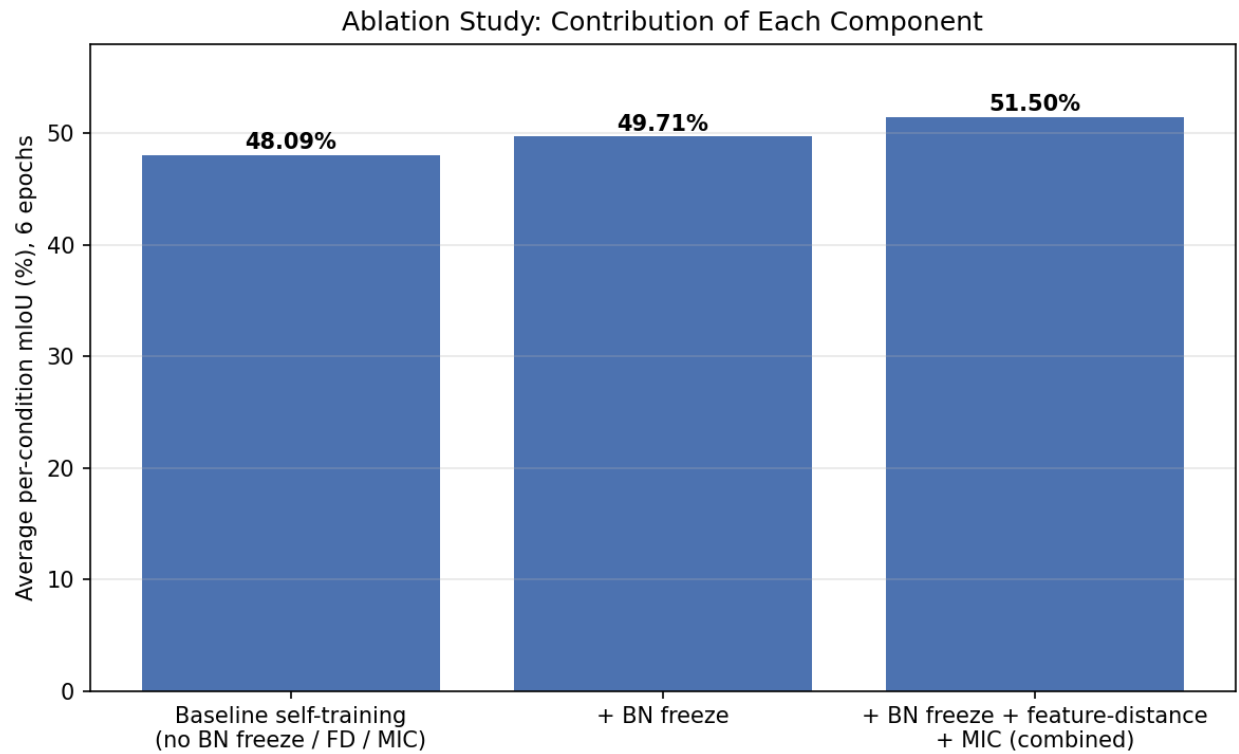


Figure 1: Ablation study — average per-condition mIoU after 6 epochs for baseline self-training, self-training + BatchNorm freeze, and the full combined configuration.

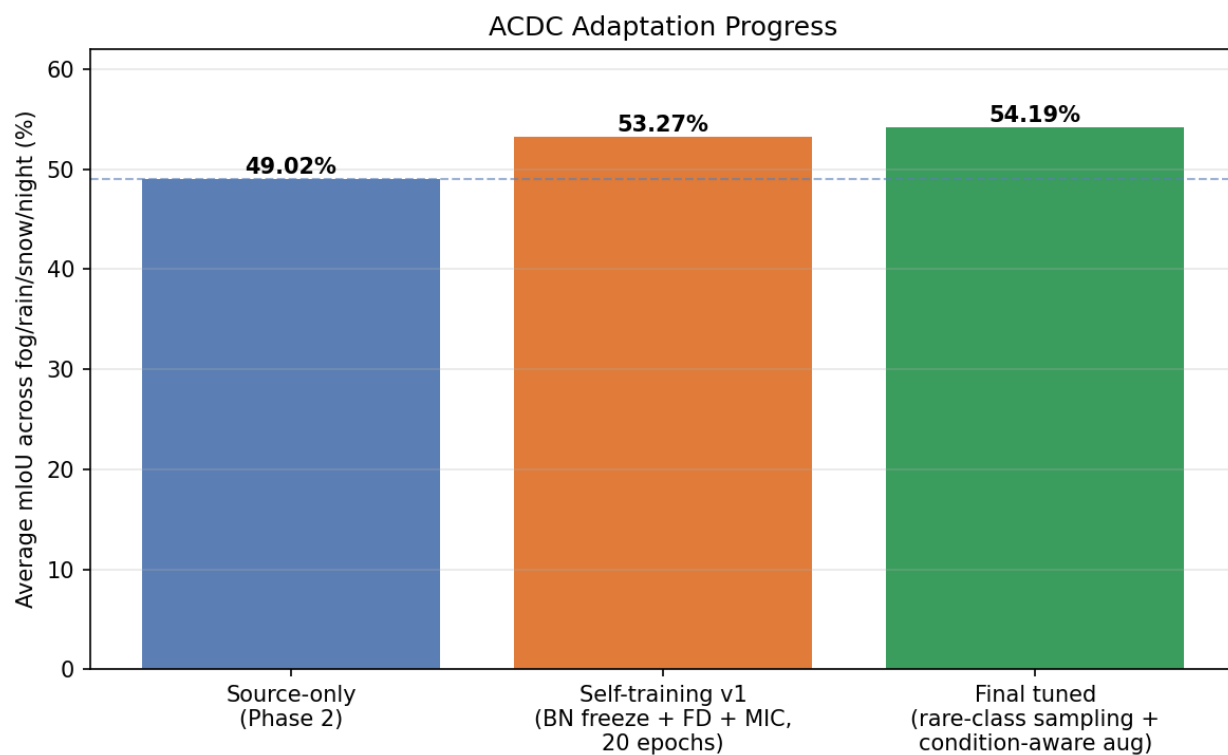


Figure 2: Average mIoU across fog/rain/snow/night at each stage of the pipeline, from the Phase 2 zero-shot baseline through the first combined self-training run to the final tuned result.

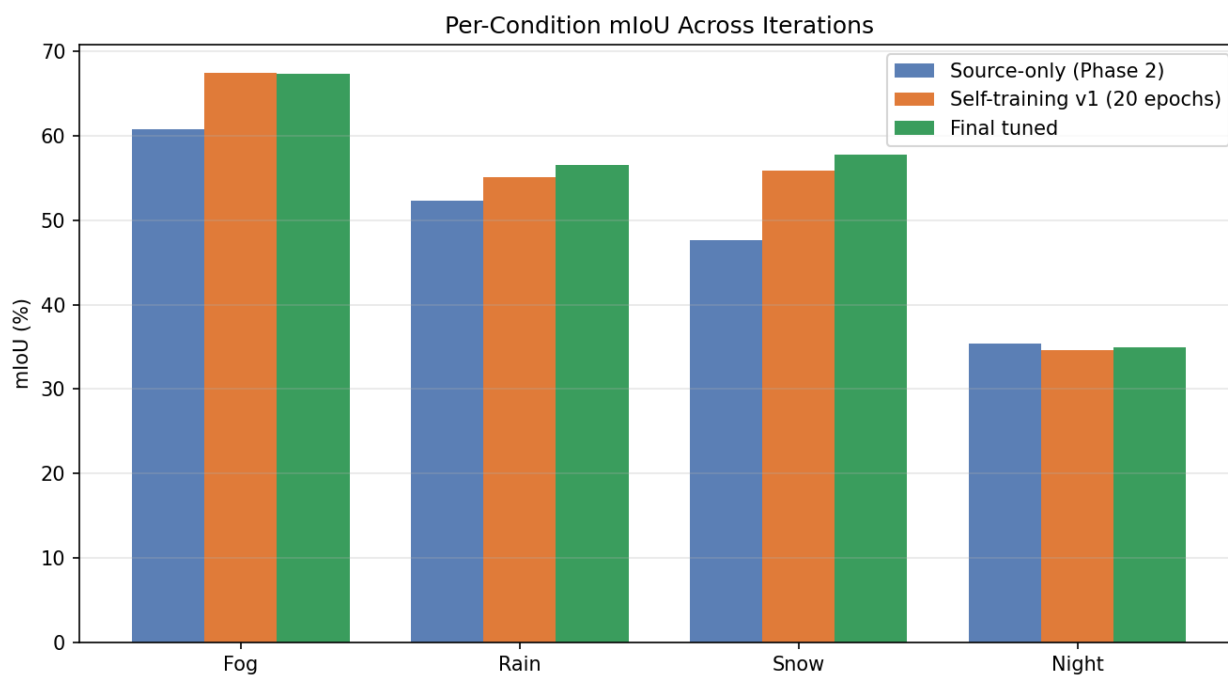


Figure 3: Per-condition mIoU at each stage of the pipeline. Night is the only condition that has not yet clearly exceeded its zero-shot baseline.

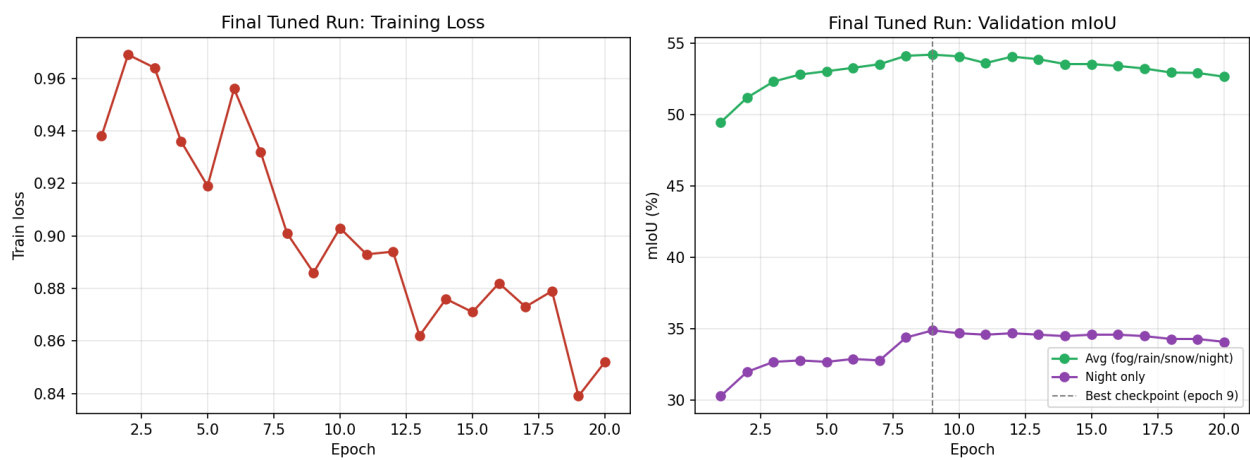


Figure 4: Final tuned run — training loss and validation mIoU (average and night-only) over 20 epochs, showing stable loss and a peak at epoch 9 (best checkpoint).

Qualitative Comparisons

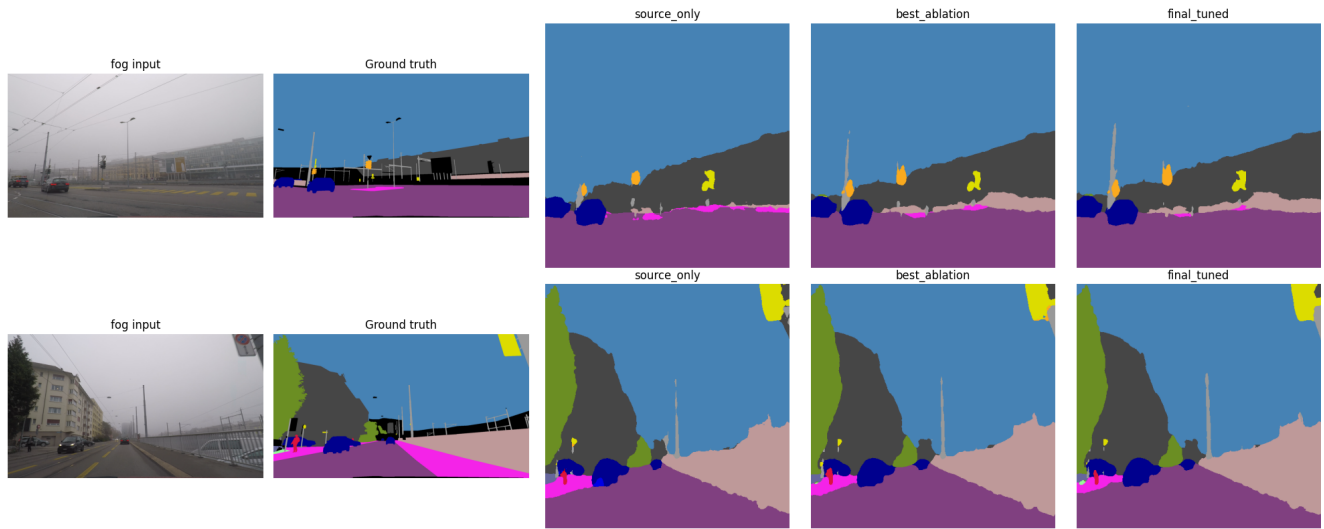


Figure: Fog — input / ground truth / source-only / best ablation / final tuned prediction, two sample crops.

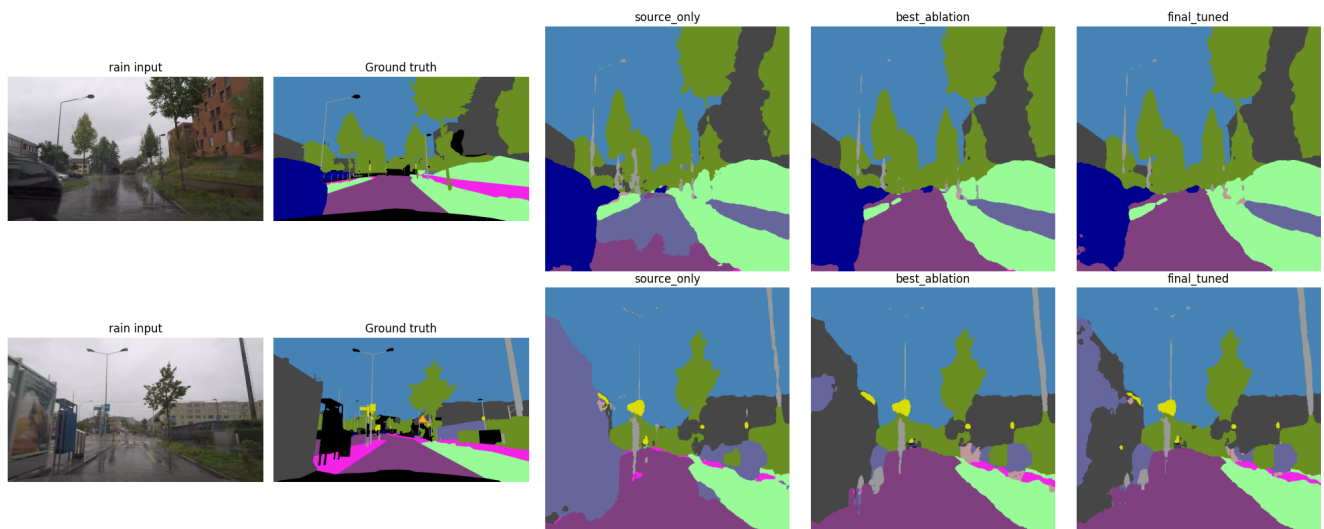


Figure: Rain — input / ground truth / source-only / best ablation / final tuned prediction, two sample crops.

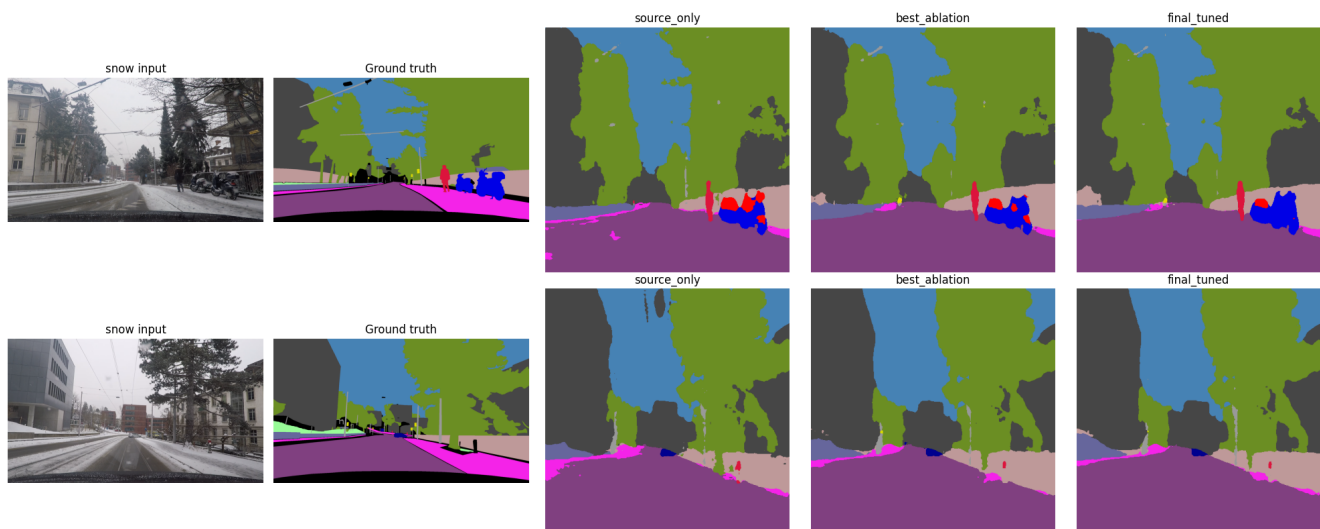


Figure: Snow — input / ground truth / source-only / best ablation / final tuned prediction, two sample crops.

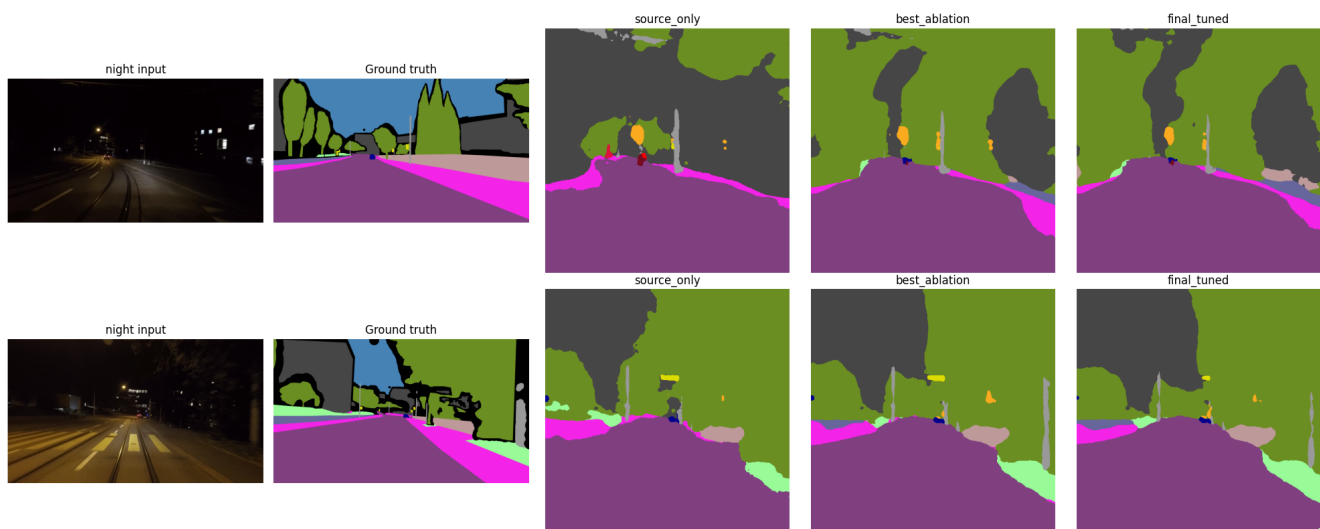


Figure: Night — input / ground truth / source-only / best ablation / final tuned prediction, two sample crops.